

Terry Nyirenda

AI ENGINEER

+27 72 921 1904 terrynyirenda123@gmail.com linkedin.com/in/terry-nyirenda-210455170 github.com/Terrytd0



PROFESSIONAL SUMMARY

AI Engineer building production-ready AI agents, multi-agent systems, Retrieval-Augmented Generation (RAG) platforms, and backend infrastructure. Experienced designing FastAPI services, LangGraph workflows, CrewAI agents, PostgreSQL and Redis-backed architectures, automated testing, CI/CD and LLM integrations. 2+ years evaluating frontier language models at Invisible Technologies combined with independent development of AI systems that solve real business problems.

PROFESSIONAL HIGHLIGHTS

- Independently built several end-to-end AI automation and retrieval platforms integrating LLMs, REST APIs, workflow orchestration, and backend services — including a legal RAG platform with automated evaluation and CI quality gates.
- 2+ years evaluating and improving frontier language models at Invisible Technologies.
- Combines AI integration, backend engineering, and business process automation across independent projects.
- Master's degree in Business Mathematics and Informatics — a strong quantitative foundation for AI system design.

TECHNICAL SKILLS

AI Engineering

- LangGraph
- CrewAI
- OpenAI API
- RAG / Hybrid Retrieval
- Prompt Engineering
- LLM Evaluation
- RAGAS
- Claude Code
- OpenCode

Infrastructure

- PostgreSQL
- MongoDB
- Qdrant
- Elasticsearch
- Redis
- Docker Compose
- GitHub Actions
- SQLAlchemy

Development & Testing

- Git / GitHub
- Automated Testing (Pytest)
- mypy
- Ruff
- Error Handling & Retry Logic
- uv
- Postman

Automation & Orchestration

- n8n
- Workflow Automation
- Airtable
- Slack Integrations
- Email Automation

Data & Analysis

- Excel
- Statistical Analysis
- Data Quality Assurance

Backend

- FastAPI
- REST APIs
- Pydantic

Programming Languages

- Python
- C++
- SQL

Authentication

- OAuth2
- JWT

PROJECTS

Independent projects, built and documented outside of employment. More projects at github.com/Terrytd0, including the Lead Intelligence Automation Platform (n8n + OpenAI-powered lead enrichment and CRM automation).

LexRAG

github.com/Terrytd0/LexRag

Independent Project — Legal contract & case-law retrieval-augmented generation platform

- Built a legal RAG platform that ingests PDF contracts and case-law documents and answers questions with traceable citations, refusing to answer when evidence is insufficient.
- Implemented hybrid retrieval — parallel Qdrant dense-vector and Elasticsearch BM25 search merged via Reciprocal Rank Fusion — refined by a cross-encoder reranker, achieving 100% recall@10 and 88% precision@5.
- Built a RAGAS-based evaluation harness (faithfulness, context precision/recall, answer relevancy) as an automated CI quality gate, using it to validate a 53% latency reduction (56s → 26s) and a refusal-accuracy prompt fix.
- Built citation-grounded generation with refusal logic and a FastAPI REST API (upload, query, document browse/delete) with SHA-256 duplicate detection and document-scoped query filtering, fully Dockerized (API, MongoDB, Qdrant, Elasticsearch).

Technologies: Python, FastAPI, MongoDB, Qdrant, Elasticsearch, RAGAS, Docker Compose, OpenAI-compatible LLM API, uv, Pytest

SupportOps AI Production-style multi-agent customer support platform - Designed and built a multi-agent customer support platform using LangGraph, CrewAI, FastAPI, PostgreSQL, Redis and Docker Compose. - Designed a policy-gated workflow that classifies customer requests, routes them to specialist agents, and escalates high-risk cases for supervisor approval. - Implemented JWT authentication, Redis-backed rate limiting, audit logging, idempotent request handling, and knowledge-base retrieval. - Built an automated test suite and GitHub Actions CI pipeline (linting, type checking, tests) covering API endpoints, graph execution, policies, and integration workflows. Technologies: Python, FastAPI, LangGraph, CrewAI, PostgreSQL, Redis, Docker Compose, GitHub Actions, JWT, SQLAlchemy, Pytest, Ruff, mypy Finance Intelligence Platform Independent Project — End-to-end AI-powered financial data automation platform - Designed and built a FastAPI backend exposing REST endpoints for financial data ingestion and AI-driven analysis. - Built a seven-stage automated pipeline — ingestion, validation, KPI calculation, AI-driven anomaly detection, executive reporting, notifications, and audit logging. - Integrated the OpenAI API for anomaly detection and executive report generation, orchestrated via n8n workflow automation, with automated tests and retry logic. Technologies: Python, FastAPI, OpenAI API, n8n, Airtable, Slack API, Pytest, Claude Code	github.com/Terrytd0/Supportops-AI Demo Video github.com/Terrytd0/Finance-Intelligence-Platform Demo Video
PROFESSIONAL EXPERIENCE Advanced AI Data Trainer Invisible Technologies - Collaborate on evaluation of frontier language models using structured prompting, reasoning analysis and quality assessment across complex real-world tasks. - Conducted systematic error analysis on AI-generated outputs, identifying and documenting failure patterns to drive model quality improvements. - Applied prompt engineering techniques to simulate realistic, complex user scenarios and expand evaluation coverage of edge cases. BMI Intern Absa Group - Performed quantitative analysis and predictive modeling to support stress testing and scenario analysis for enterprise risk assessment. - Conducted research and statistical analysis supporting model development, validation, and business risk decision-making. - Contributed to predictive modeling initiatives within structured, data-driven business analytics workflows.	May 2024 – Present Remote Aug 2022 – Dec 2022 Johannesburg, South Africa
CERTIFICATIONS IBM Full Stack Software Developer Professional Certificate Salesforce Certified Administrator	Jun 2025 May 2026
EDUCATION Master's Degree, Business Mathematics and Informatics North-West University Honours Degree, Quantitative Risk Management North-West University Bachelor of Science, Actuarial Science North-West University	Jan 2022 – Mar 2023 Potchefstroom, South Africa Jan 2021 – Dec 2021 Potchefstroom, South Africa Jan 2018 – Dec 2020 Potchefstroom, South Africa